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Successful Deployment of the World's First Density-Based Inflow Control System for Improved Well Performance

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Abstract

Inflow control technologies have helped oil and gas companies maximize oil recovery and minimize water production by balancing production across the entire completion. To further minimize water production and maximize oil recovery, the industry started utilizing autonomous inflow control devices (ICDs) and systems that can autonomously sense the incoming fluid properties and choke if the properties match that of water and freely produce if the properties match oil. However, all autonomous inflow control devices (AICDs) available in the market to date require a viscosity contrast between oil and water to be able to choke formation water. Light oil reservoirs, which are abundant around the world, do not have sufficient viscosity contrast between oil and water. In fact, some reservoirs have ultra-light oil where the oil viscosity is even lower than water. In those reservoirs a new technology was required to operate on a fluid property other than viscosity. This paper presents the design concept, qualification and successful deployment of the world's first autonomous inflow control system that differentiates between the downhole oil and water solely off the fluid's density.

Introduction

In the late 1980s and 1990s the oil industry started to move towards drilling horizontal wells to increase reservoir contact thereby maximizing oil production. It quickly became evident that in heterogenous formations, zones that have higher permeability produce at a much higher rate and accelerate the process of pulling water line upwards in that zone. This creates what the industry has coined as water coning as shown in the left picture of Figure 1 which is a highly permeable zone at heel. This led to high water production early in the well's life which significantly reduces the overall oil recovery particularly in zones where the permeability is relatively lower.

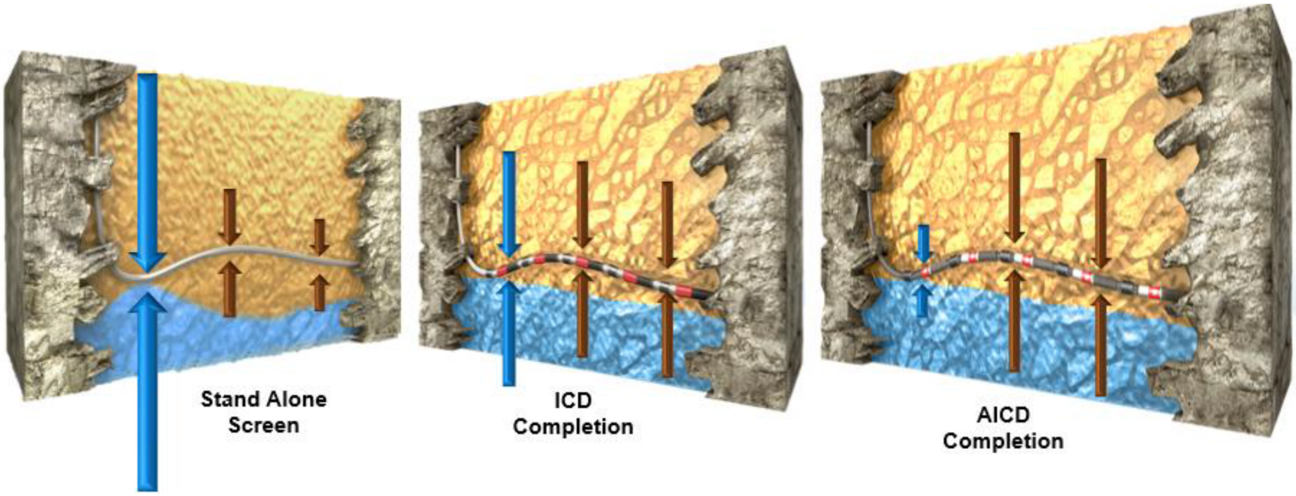

Figure 1—Flow behavior comparison between stand-alone screens, ICDs and AICDs. Brown arrows indicate oil flow, and blue arrows indicate water flow

To address this issue the industry introduced the first inflow control devices in the 1990s *(Brekke & Lien, 1994)*. ICDs are designed to have an adjustable flow area which is achieved by configuring nozzle or tube sizes inside the tool. In zones where permeability is high the tool is configured to have a smaller flow area and vice versa in zones with lower permeability. The goal is to achieve balanced production across the entire producing horizontal section of the well to assist in controlling water coning. This is done by logging the well to measure the permeability along the horizontal reservoir section then configuring and deploying the ICD completion accordingly.

Although ICDs help delay water production, the water will breakthrough sooner or later. Therefore, to further minimize water production and maximize oil recovery the industry introduced autonomous inflow control devices in the 2010s (Mathiesen et al., 2011). AICDs are designed to autonomously detect the incoming fluid from the reservoir then autonomously choke it if it is water or freely produce if it is oil. This allows autonomous choking in zones that are producing water along the horizontal reservoir section. This in turn means more production in zones that are still producing oil.

Over the years several designs were introduced to the market, but they all required a minimum viscosity difference between oil and water. This provided a good solution for reservoirs where the oil has a high enough viscosity compared to formation water. However, there are many reservoirs that have light better-quality oil where the viscosity of oil is very close to that of water or even in many cases the oil has a lower viscosity than water. In those reservoirs all existing AICD technologies would not be able to perform their function.

Since there were no available autonomous inflow control systems in the market that would work in light oil reservoirs a new technology was required. This paper introduces the design concept, qualification and successful deployment of the world's first autonomous inflow control system that operates solely on the density contrast of the downhole fluids.

Technology Overview and Key Features

There are five main subsystems that make up the density-based inflow control system which are shown in Figure 2. These are

- Screen/Solids barrier
- Centrifugal Fluid Selector (CFS)

- Production Valve
- Washpipe Free Feature
- Sliding Side Door (SSD)

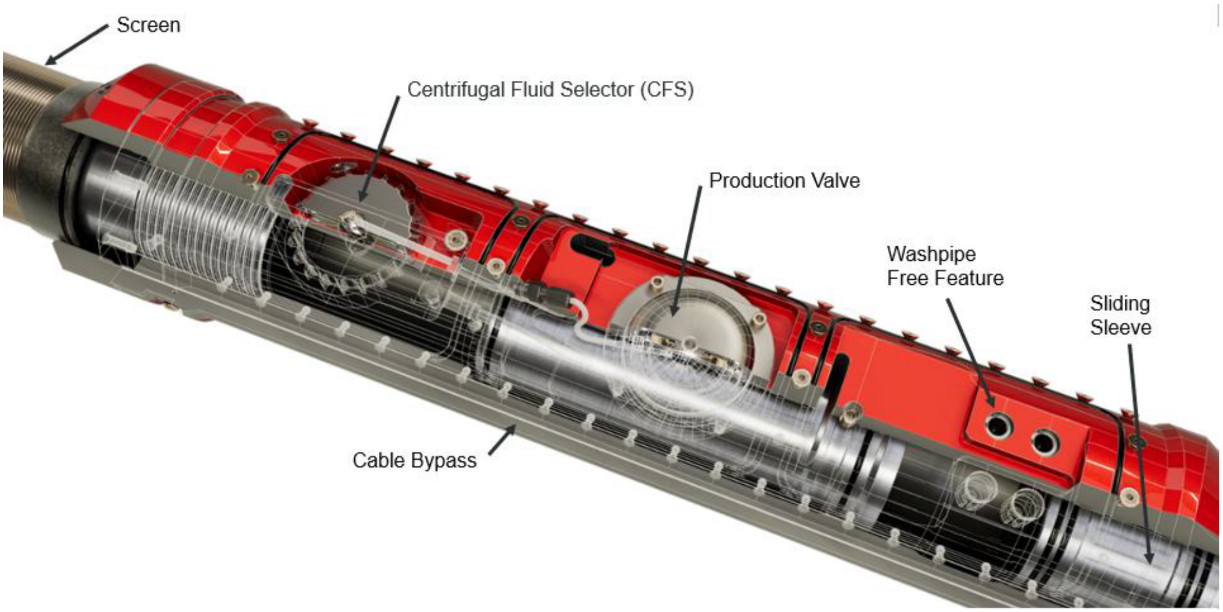

Figure 2—Different components of the density based autonomous inflow control system.

All the flow incoming from the formation must first pass through the screen to filter any particulates out of the fluid. After passing through the screen the flow splits into two paths, one going through the CFS and the other going through the production valve. The CFS receives a little amount of the flow, whereas the production valve receives a much larger percentage (up to 10:1). The CFS identifies whether the fluid density is above or below a predefined threshold. When it detects a lighter fluid — such as oil — the production valve remains in its default open state, allowing a high flow rate through the system. However, if the CFS senses a heavier fluid, like water, it triggers a pressure response that closes the production valve, which significantly restricts the flow through the tool. The washpipe free feature acts like a check valve by not allowing fluid to flow from tubing to annulus while allowing free flow from annulus to tubing. This design supports the deployment of mechanical packers with the tubing and ensures circulation from the toe and not through the screens while running in the hole. Additionally, a built-in SSD (Sliding Side Door) can be mechanically shifted at any point to completely block flow through the device if needed.

Screen/Solids Barrier

The screen's main function is to remove any particulates from the fluid. This is particularly critical in sand formations due to the high erosivity of sand particles. This is not only critical to protect the tool from erosion, but the entire tubing string and surface equipment. The screen of the tool can be customized to fit different applications. The screen can be customized to have different mesh sizes, different lengths, different materials, and different types. For instance, mesh screens have been found to be less prone to plugging by mud and are better suited for aperture sizes below 200 microns compared to wire wrap screens (Furgier et al., 2013). Mesh screens have a larger flow area compared to wire wrap and therefore are more suited for applications where screen erosion might be a concern.

Centrifugal Fluid Selector

The Centrifugal Fluid Selector (CFS) functions as a mechanical density sensor by generating artificial gravity. Inside the CFS are two specially designed floats with densities that fall between those of oil and water. These floats move radially — either inward or outward — based on the density of the surrounding formation fluid, rather than moving vertically like traditional gravity-based systems. This radial movement is driven by centrifugal force, created as the CFS continuously spins using a portion of the production fluid. This artificial gravity greatly enhances buoyant forces acting on the floats, rendering Earth's gravity irrelevant and eliminating the need for specific downhole orientation. The motion of the floats actuates a pilot valve, which in turn controls the production valve. Powered entirely by formation fluids, the CFS remains active throughout the well's life, automatically adjusting to changes in fluid composition over time.

Production Valve

The production valve regulates the primary flow through the tool. When it is closed, fluid flow is restricted to a small volume flow rate passing through the CFS. A pilot line connected to the CFS governs the production valve position. When water is present, the CFS redirects flow into the pilot line, generating pressure that forces the production valve to close. In contrast, when oil flows through the tool, the CFS blocks the pilot line, relieving the pressure and allowing the valve to return to its default open state. With the valve open, the total flow rate through the tool can be up to ten times higher than when the valve is closed, achieving a 10:1 oil-to-water flow ratio that enhances oil recovery while minimizing water production. Built from highly erosion-resistant materials, the production valve can handle flow rates of up to 700 barrels per day at a low differential pressure of just 150 psi. The tool's flow rate range can be configured on-site before deployment based on updated logging or simulation data. Multiple tools can be installed within a single well interval to fine-tune production rates and balance the inflow profile effectively.

Washpipe Free Feature

The washpipe-free feature functions as a check valve, allowing fluid to flow freely from the annulus into the tubing while blocking any reverse flow from the tubing to the annulus. One key benefit of this design is that it directs circulated fluids to exit through the bottom of the tubing string and return up the annulus, ensuring efficient displacement of the existing fluid. For instance, when using a mud breaker, this configuration guarantees full contact with the filter cake throughout the annulus. Additionally, the washpipe-free feature is built to withstand tubing pressures of up to 5000 psi, which can be used to set mechanical packers. This removes the need for washpipe to isolate the screens, offering substantial savings in both time and operational costs (Wang et al., 2021).

Sliding Side Door (SSD)

The density based autonomous inflow control system includes a sliding side door (SSD) that is usually deployed in the open position. Flow from specific joints can be stopped by mechanically shifting the SSD(s) to the closed position via an intervention. This shifting can be performed using either a hydraulically operated coiled tubing shifter or a wireline stroker. Both methods enable precise, selective actuation of targeted sleeves without affecting the position of the others.

Additional Features

Cable Bypass. The density-based autonomous inflow control system is equipped with a cable bypass that enables control line feed-through, as illustrated in Figure 2. This allows electric or fiber optic cables to be deployed alongside the tool for monitoring well parameters such as pressure and temperature. The bypass channel is designed to accommodate a 22x11 mm flatpack.

Field Adjustable Nozzle. The density based autonomous inflow control system tool includes a field-adjustable nozzle, allowing for last-minute adjustments to the effective flow area before the completion equipment is deployed. This might be needed to help balance the production flow across zones to help alleviate water coning as described earlier. Changing the nozzle, which modifies the tool's flow performance (flow rate versus pressure drop), can be done quickly and easily at the rig site — typically within minutes. The nozzle size influences only oil production, leaving the water production curve unaffected. In addition to preventing water coning, selecting the appropriate nozzle is crucial to match specific well conditions and keep the tool operating within its optimal pressure range. A dedicated calculator is available to generate flow performance curves, as illustrated in Figure 3, using just three inputs: in-situ oil density, water density, and the selected nozzle size. As shown in Figure 3, oil-to-water ratios as high as 10:1 can be achieved using the extra-large nozzle.

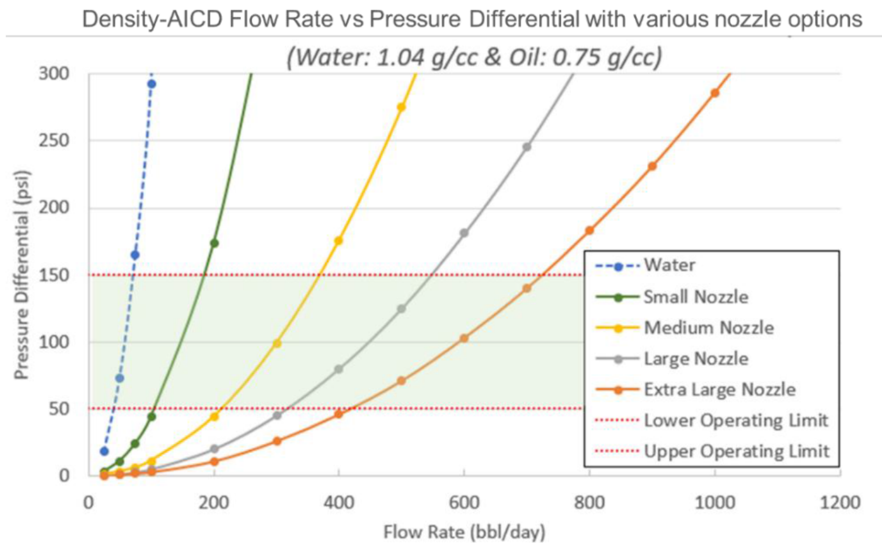

Figure 3—Flow rate vs pressure drop of different nozzle sizes. The water curve is not affected by the nozzle size.

Qualification

The performance of the density-based autonomous inflow control system was validated through a combination of numerical simulations, small-scale component tests, and a full-scale qualification program. Key tests assessed both component-level and full-system reliability, including cycle and erosion testing of the production valve, bearing erosion, environmental testing (TM0296-2014 at 3,000 psi and 125°C), wash-pipe-free pressure tests, and full system erosion and flow performance evaluations for single and multi-phase conditions.

To ensure mechanical integrity during deployment, finite element analysis (FEA) and physical testing were conducted. FEA simulated conditions like dog-leg severity, burst and collapse of sliding sleeves, and tensile loading—showing no plastic deformation. Physical testing included bolt-on cover plate burst/ collapse tests and flow erosion trials inspired by API 19ICD standards. Figure 4 below provides an overview of the testing performed.

A baseline flow loop test confirmed expected single-phase flow behavior and water cut performance in multiphase flow. The tool then underwent an aggressive erosion test using high sand concentration (exceeding field conditions), with approximately 3,595 lbs of sand passed through over 31 hours. Post-test flow results showed minimal impact, with only a 0.4% change in pressure drop, confirming the tool's durability and functionality after erosion exposure.

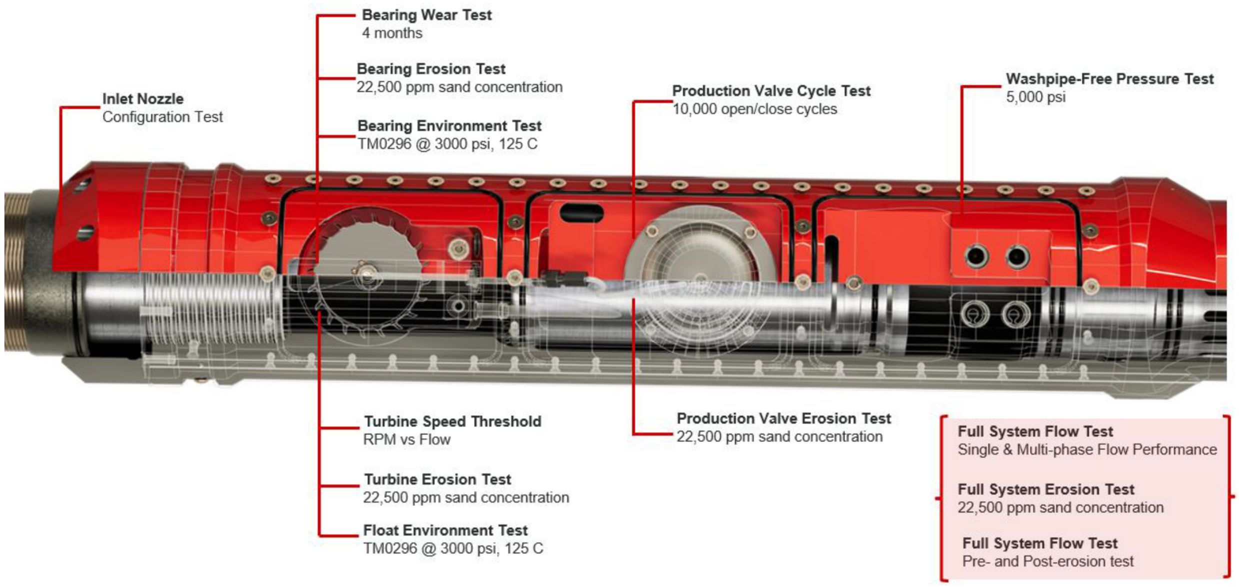

Figure 4—Overview of qualification tests performed.

Full Scale Test 1

The density based autonomous inflow control system was tested under both single-phase and multiphase flow conditions to assess its performance in realistic downhole environments. The primary goal was to evaluate the tool's behavior with fluids that closely replicate actual downhole viscosities and densities. Exxsol D60 simulated light crude oil, while NaCl brine was mixed to match field brine characteristics. The test program included flow characterization testing followed by erosion testing, then repeating the flow characterization testing to prove that the tool's functionality was not affected by the erosion. Table 1 below summarizes the test parameters.

Table 1—Test parameters of first qualification test.

Test temperature	57°C (135°F)
Test pressure	69 bar (1000 psi)
Differential pressure for single phase	3.5, 5, 7, 10, and 20 bar
Differential pressure for multiphase	7 bar
Multiphase water cut %	0-100, 100-0
Differential pressure for erosion	7 bar
Total sand passed	3,595 lbs
Test fluid densities at test conditions	Oil: 780 kg/m^3 Brine: 1044 kg/m^3

In single-phase testing, the tool showed its ability to autonomously restrict brine flow and establish the flow vs. differential pressure profile for both large and small production valve pre-choke nozzles. Two different nozzle sizes were characterized in the flow and multiphase testing. Multiphase testing determined the water cut percentages at which the valve would close and reopen. A third-party lab conducted all validation tests.

An erosion test was followed to assess performance degradation after prolonged sand slurry exposure. The production valve, normally closed during water production, was kept open by plugging the pilot line, allowing sand to flow through the entire system. Post-erosion flow tests and inspection showed minimal wear: a maximum mass change of only 0.10% and a pressure differential change of 0.4%. A total of 3,595 lbs of sand passed through the tool—exceeding the API 19ICD requirement of 3,124 lbs—helping illustrate the system's durability and consistent functionality after erosion exposure. More details on testing can be found in El Mallawany et al., 2023.

Full Scale Test 2

Flow performance testing of the density based autonomous inflow control system was conducted under downhole-like temperature and pressure conditions using crude oil from the Ivar Aasen field in Norway. These tests involved two different adjustable oil production nozzles: Nozzle A with a 3.8 mm (0.15") internal diameter and Nozzle B with a 6.6 mm (0.26") internal diameter. The same tool sample was used throughout the testing, with only the nozzle swapped between runs. Table 2 below summarizes the test parameters.

Table 2—Test parameters of second qualification test.

Test temperature	100°C (212°F)
Test pressure	200 bar (2900 psi)
Differential pressure for single phase	3.5, 4, 5, 7, 10, 15 and 20 bar
Differential pressure for multiphase	3.5, 5, 7 and 10 bar
Multiphase water cut %	0-100, 100-0
Test fluid densities at test conditions	Crude Oil: 758 kg/m^3 Synthetic formation water: 962 kg/m^3 Natural gas: 117 kg/m^3

With Nozzle A, the density based autonomous inflow control system performed similarly to a nozzle-type ICD, allowing unrestricted oil flow while significantly reducing flow once water was detected. The nozzle had an effective oil flow performance comparable to a 4.2 mm ID nozzle, delivering a 5.3:1 oil-to-water ratio at 7 bar differential pressure.

Multiphase testing with Nozzle A assessed tool behavior across four differential pressures. The tool closed at water cut values ranging from 86% to 100%, with slightly lower closing thresholds as pressure increased. Re-opening occurred between 35% and 43% water cut, also varying slightly with pressure. The tool's hysteresis ensured that it maintained a stable state rather than oscillating between open and closed.

With Nozzle B, the tool provided greater oil flow capacity, acting like a 5.8 mm ID nozzle and producing a 10.3:1 oil-to-water ratio at 7 bar differential pressure. The flow behavior aligned closely with predictive models and showed substantially higher restriction to water compared to an equivalent nozzle ICD.

Multiphase testing with Nozzle B followed a similar approach, including pressure differentials across the full planned range. The tool closed at water cuts between 94% and 100%, again with earlier closure at higher pressures. It re-opened between 40% and 42% water cut, mirroring the stable and consistent performance observed with Nozzle A. More details on testing can be found in Corona et al., 2024.

Simulation Modeling

Comprehensive simulation modeling was performed to capture the value of the technology, especially in terms of improvement in water cut and oil production. The simulations showed the ability of the technology to effectively choke water production from the watered-out zones without any surface command or intervention. The autonomous behavior of the tool was the key to controlling water production downhole. Multiple simulation scenarios were performed to predict the behavior of the tool with varying completion designs to optimize the completions. Open-hole completions were referred to as a base case to quantify the improvement in water cut percentage. In addition, conventional ICD scenario was also run to compare to the autonomous Density-based inflow control system.

The open-hole section was compartmentalized in 5 zones based on well logging and PLT data. The downhole fluids densities were also incorporated in the simulation. A total of 13 Density-based inflow control systems were incorporated in the completion design with a nozzle size of 0.20". The simulation results showed that the autonomous Density-based inflow control system improved the water-cut percentage by ~25% as compared to conventional ICD technology. This significant improvement was observed due to the autonomous Density-based inflow control system restricting water in the high-water saturation zones.

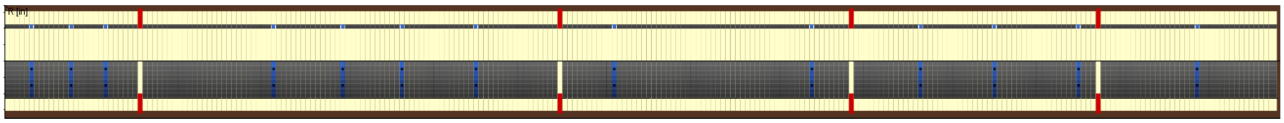

Figure 5—Completion Design for Simulation modeling

It was observed that the Density-based inflow control system effectively reacts to watered-out zones by autonomously restricting the water downhole. The below Figure 6 shows the cumulative water production of open-hole scenario, ICD completion scenario and Density-based inflow control system. The Density-based inflow control system scenario results in the lowest cumulative water production, while the open-hole scenario showed the highest cumulative water production.

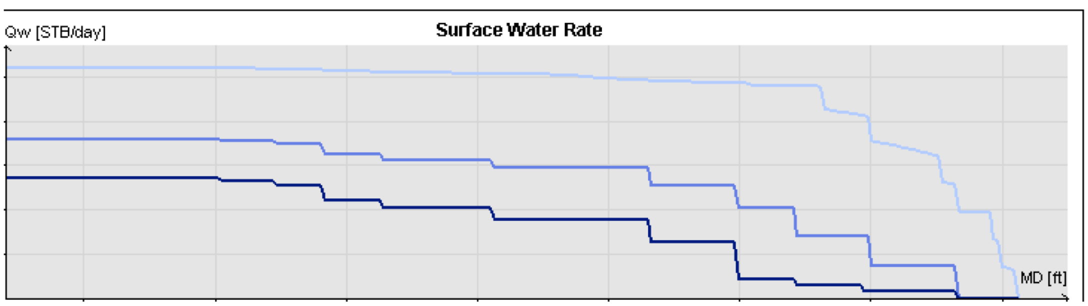

Figure 6—Cumulative Water Production for various scenarios

The below Figure 7 shows the cumulative oil production of open-hole scenario, ICD completion scenario and Density-based inflow control system. The Density-based inflow control system scenario results in the highest cumulative oil production, while the open-hole scenario showed the lowest cumulative oil production.

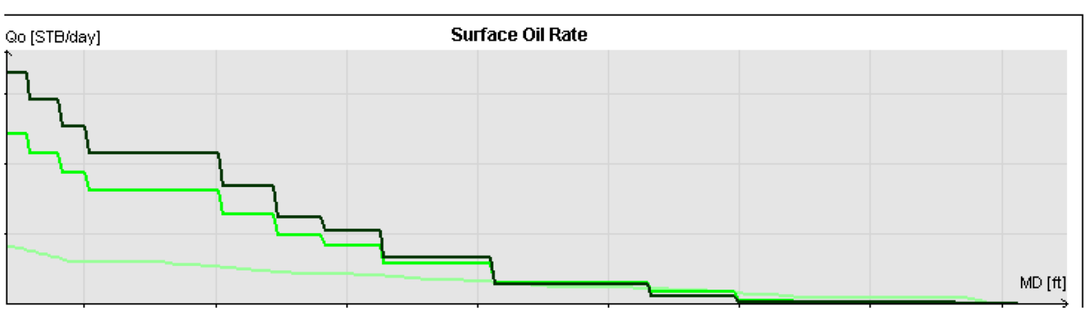

Figure 7—Cumulative Oil Production for various scenarios

Field Deployment

The Density-based inflow control system was thoroughly inspected and tested in the workshop before proceeding with the deployment plan. Once the design was finalized based on the simulation modeling results, the Density-based inflow control system was successfully deployed in an onshore oil well as part of the lower completions to reduce water production. The length of the completion string was approximately 4,000 ft that was deployed in a highly deviated well. Density-based inflow control system was deployed at the target depths without any issues or concerns. All the joints were run with the SSD in open position, to be shifted to closed position in the future if needed. Cleanout and reaming trips were performed before running in hole with the system to ensure a smooth run with the completion string. A holistic contingency plan was put in place to counter any unforeseen and unfavorable situation during operations.

After reaching TD, OBM was displaced with Brine and mud cake removal fluid was pumped. The packers and liner hanger were then hydraulically set by pressuring up the tubing after dropping a ball to close the bottom of the completion string. All the jewelry was hydraulically set in the same pressure cycle. The Density-based inflow control system allowed pressuring up the tubing without any requirement of wash-pipe or inner string. It held an internal pressure of 3,500 psi to set the open-hole packers and liner hanger.

After setting the lower completion in place, the running tool was pulled out of hole and upper completions was deployed as per the design. The Density-based inflow control system was successfully deployed in accordance with the specified program and design. Further evaluation will be conducted in the future during the flowback and production phase of the well, including PLT logging. This would confirm the downhole behavior of the technology at the compartment level.

Conclusions

The successful deployment of the world's first density-based autonomous inflow control system marks a significant advancement in enhancing well performance in light oil reservoirs. By leveraging fluid density rather than viscosity, this innovative technology provides a game-changing solution for minimizing water production and maximizing oil recovery. Extensive qualification testing, simulation modeling, and field deployment have proven its reliability, durability, and effectiveness in real-world conditions. The system's ability to autonomously adapt to changing downhole fluid compositions ensures sustained production optimization throughout the well's life. As the industry continues to seek improved methods for inflow control, the density-based approach presents a viable alternative to conventional AICD technologies, paving the way for broader adoption in diverse reservoir conditions.

References

Brekke, K., & Lien, S. C. (1994). New, Simple Completion Methods for Horizontal Wells Improve Production Performance in High-Permeability Thin Oil Zones. *SPE Drilling & Completion*, **9**(03), 205–209. https://doi.org/10.2118/24762-PA

Corona, G., Greci, S., Novelen, R., Warren, C., & Langaas, K. (2024, April 17). Performance and Modelling of a Density-Based Autonomous Inflow Control Device. SPE Norway Subsurface Conference. https://doi.org/10.2118/218456-MS

El Mallawany, I., McChesney, R., Greci, S., Novelen, R., Corona, G., Warren, C., Asthana, P., & Bukhamseen, A. (2023, October 2). Qualification of an Emerging Class of AICDs with Artificial Gravity: The Density-AICD Technology. *ADIPEC*. https://doi.org/10.2118/216652-MS

Furgier, J.-N., Viguerie, B., Aubry, E., & Rivet, P. (2013, June 5). Stand Alone Screens: What Key Parameters are Really Important for a Successful Design? SPE European Formation Damage Conference & Exhibition. https://doi.org/10.2118/165170-MS

Mathiesen, V., Aakre, H., Werswick, B., & Elseth, G. (2011, September 6). The Autonomous RCP Valve - New Technology for Inflow Control In Horizontal Wells. SPE Offshore Europe Oil and Gas Conference and Exhibition. https://doi.org/10.2118/145737-MS

Wang, Z., Newton, D., Qureshi, A., Uchiyama, Y., Corona, G., & Ashraf, H. (2021, September 15). Innovative Washpipe Free Circulating ICD Delivers Reduced Well Construction Costs. SPE Annual Technical Conference and Exhibition. https://doi.org/10.2118/206209-MS